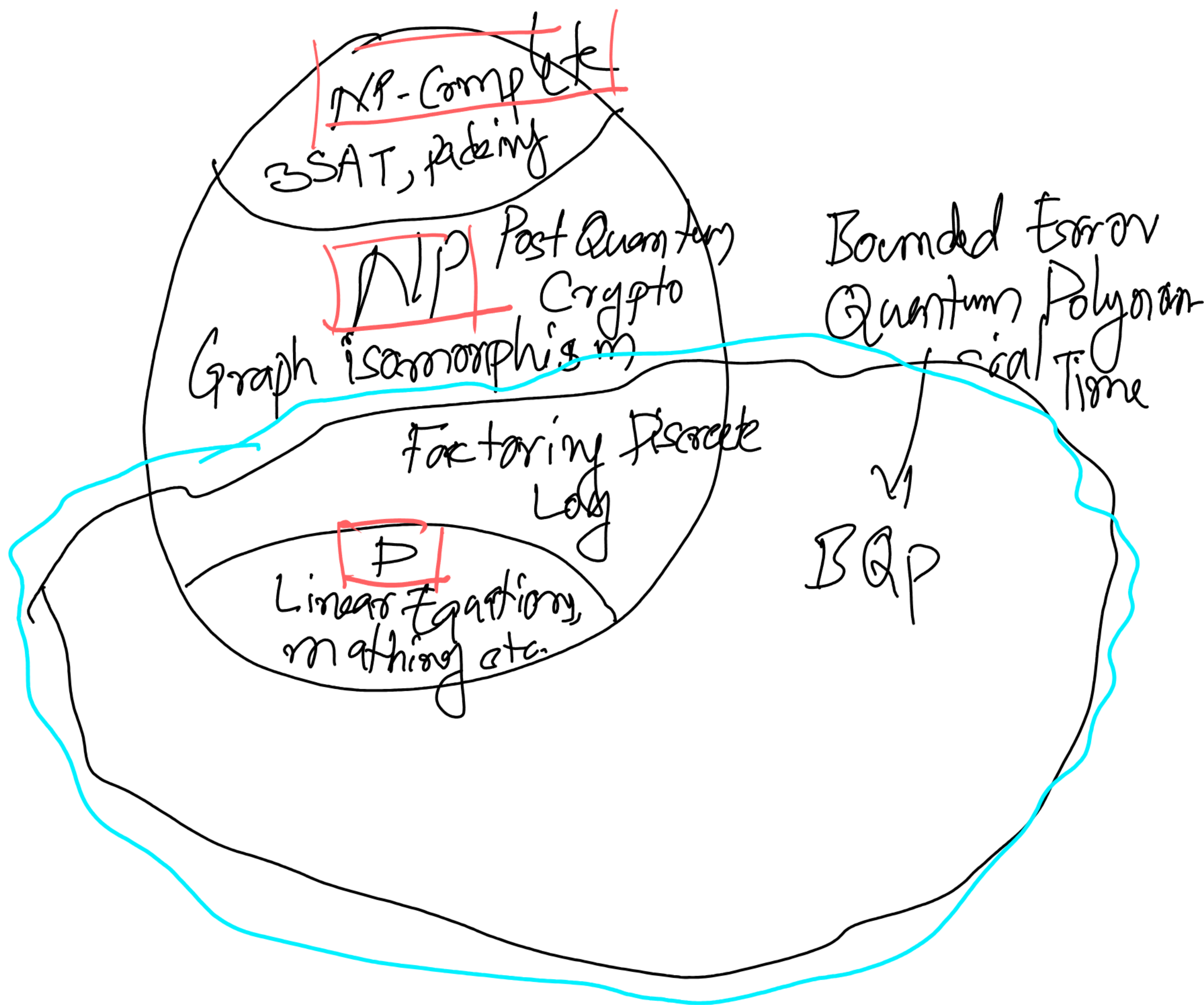




P = Problems that are solvable in polynomial time. using a classical Computer

NP = All the problems where a classical Computer can at least efficiently verify the answer

P - sup e. or various cross ed
problem whether $P \equiv NP$

BQP is contained in NP
- classical computer can't even check
the answer.

Quantum Supremacy

103 superc^{conducting} qbits. classical comp
- iter takes 10^{25} years. to . . .

Fermi-Hubbard model

Out of Time Order Correlators (OTOC)
↳ Comparison/Check - Second QC
- Physics experiment

Many - Worlds Interpretation
(Everett 1957)

but we call 'Measurement of a state $|\psi\rangle$ '
is just us and our environment getting
entangled with $|\psi\rangle$, as the word splits
into equally-real branches.

$$(\alpha|0\rangle + \beta|1\rangle)|\text{You}\rangle$$

$$\rightarrow \alpha|0\rangle|\text{You}_0\rangle + \beta|1\rangle|\text{You}_1\rangle$$

Niels Bohr hated it and on asked
to censor, he became nuclear weapon
strategist.

Main Alternatives to Many Worlds

- Bohm-deBroglie (first amplitude + "real branch")
 - New Physics (e.g. dynamical collapse mechanism)
 - Copenhagen ("learn to stop asking the question!")
-

David Deutsch — 1997 challenge

When Shor's algorithm has factorized a number, using 10^{500} or so times the computational resources that can be seen to be present, where was the number factorized? There are only 10^{80} atoms in the entire visible universe.

philosophical problem

Exponentiality of the amplitude vector

"Is anybody home in ^{the} other branch?"

— Quantum Computing is silent.

Many worlds collided — that's QM

Complexity - Theoretic Occam's Razor
- Zero's finite-time Goldbach Decider

$$4 = 2 + 2$$

Goldbach's Conjecture - Every even
number four or greater can be written
as sum of two prime numbers.
(unsolved problem)